# PROJECT PROPOSAL:

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I have decided to do a Project where I extract and analyse the formants F1 and F2 of different Spanish accents to be able to build multiple vocal spaces. My main idea is to use Mozilla Common Voice (https://mozilladatacollective.com/datasets/cmn4z1n52000knv07h01532dd ) because it is open source, and because they have a significant amount of data to work with.

There are many different types of accents, but I think it would be better to look for peninsular dialect variations (like North vs. South) or standard Spanish compared to Mexican. For now, I do not know how many I will actually use. I would like to apply DSP techniques to be able to automatize the phonetic analysis of dialect variations.

Following the course schedule, I am defining my theme in April to start working on the data. From here, my plan for the project is:

- May - June: Work on the Python code to extract the F1 and F2 formants. I will upload regular weekly progress reports (updates) in Ilias and discuss any issues I find with the code during the lectures.
- July: I will present my results in a 20 min talk with slides. I saw the presentation talks are scheduled between July 7th and July 16th.
- October: I will finish and submit the written report of 12+ pages.

# 05.05.2026 - Project Update

## Data extraction

My main problem was that the original dataset was 50 gbs big. It took me two days and loads of trouble shooting to be able to finally download everything because my laptop is very slow, and also I didn't have enough memory in the IMS server. Also, I thought that if I wanted to process all the data in the dataset (which was almost 2 million datapoints) I would run out memory before I could even achieve anything.

The fix was to first extract everything into my desktop from the zip file and then I built a small snippet of code ("preprocesador.py") which reads "validated.tsv" to extract randomly 100 audios from the north and the south variants of Spanish each. This way I could test that everything worked before trying it with the complete dataset.

## Extractor

I built another small extractor ("extractor.py") which uses Praat to extract f1 and f2 (for now) from a test .wav file of me saying /a/. It calculates the whole energy curve from the audio and it looks for the maximum intensity peak. My original code cut the audio in hald and the looked for the formants, but if I was phonating a fricative I would get completely wrong data. I assumed that vowels are phonated when the vocal tract is most open, so the peak helps me get the nucleus of the vowel.

## Pipeline

I filter everything into a folder on my computer called "audios_filtrados" and the coded processes the first 200 extracted audios one by one to get the formants. The output is a file called "resultados_formantes.csv" which saves all the information that was extracted. I will attach it in the following days, because I am having troubles with the extraction.

# 14.05.2026 - Project Update

## Vowel labelling

My goal for this session was to add vowel labels (a/e/i/o/u) to the 569 extracted tokens in “results_formants_clean.csv”(291 North, 278 South). I built a script (“etiquetar_vocales.py”) and tried two approaches:

- v1**:** I used acoustic ranges (Quilis 1981) to classify them. My problem was that it labeled almost all /a/ as North and /i/ as South. I realized this is because Quilis' ranges are calibrated for Northern Spanish. Using F1/F2 to label the data creates circularity.
- v2: I tried inferring the vowel from the transcription text instead. This was too conservative (only 14 valid tokens) because the clips are full sentences. Without phonetic alignment, I don't know exactly what phoneme is sounding during the intensity peak, so this adds too much noise.

## Methodology update

The fix was to abandon individual vowel labelling via text. From here, my plan for the analysis is:

- Global analysis: Compare the North and South distributions over the whole F1/F2 space using statistical tests (Welch's, Cohen's d). The data already shows the South has a lower F1 and higher F2, which matches Andalusian articulation.
- K-means (k=5)**:** I will let the algorithm find five clusters in the data without imposing external ranges. Then I will interpret the centroids as /a/ /e/ /i/ /o/ /u/ depending on their position. This is methodologically much better.

**Next steps**

- **Next session:** Run ”analsis.py” review the plots, and interpret the k-means results.

# 25.05.2026 - Update

**25.05.2026 - Project Update**

**Bandwidth extraction**

Following feedback from Prof. Wokurek, I updated the extractor to also extract B1 and B2 (the bandwidths of F1 and F2) alongside the formants. Bandwidth measures how "sharp" or "diffuse" each formant peak is,  a narrow bandwidth means a well-articulated, tense vowel, while a wide one indicates a reduced or more relaxed articulation. This is relevant for dialect comparison because Andalusian Spanish is known for vowel reduction. The updated script is "extractor_v2.py" and the cleaning script was also updated ("limpieza_v2.py") to filter implausible bandwidth values. After re-running the full pipeline on the new computer, I ended up with 440 clean tokens (244 North, 196 South).

**Analysis results**

I ran a full analysis ("analisis_v2.py") that answered four questions from the professor:

F1, F2, B1, B2 comparison: F1 and F2 are both significantly different between dialects (as before). The most interesting new finding is B2: the South has almost three times the B2 of the North (408 Hz vs 152 Hz, Cohen's d = 1.387), which is as large an effect as F1. B1 is not significant, which means both dialects articulate vowel height similarly, the difference is in tongue backness and how diffuse that movement is.

K-means by dialect: I ran k-means separately on North and South using the elbow method and silhouette scores to find the optimal k. Both dialects came out with k=3, meaning the data organises into three broad zones (open / mid / closed) rather than five distinct vowels. This makes sense given that I don't have phonetic alignment.

Bandwidth vs no bandwidth: Adding B1/B2 to the clustering actually lowers the silhouette score (0.370 -> 0.298), which means bandwidth captures a different kind of structure than formants do, they are not redundant dimensions. This is an interesting finding for the report.

Subgroups within the South: I ran k-means freely on just the Southern data in 4D (F1/F2/B1/B2). The optimal k came out as 5, but with low silhouette scores overall (0.26–0.28), so the subgroups are not very clearly separated. Some subgroups have very high B2 values which could point to more extreme Andalusian realisations, but I would need province-level metadata to confirm this.

**Next steps**

Prepare the slides for the presentation and start structuring the written report.